

Lattice Field Theory — Exam Questions with Solutions

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Each chapter below has three questions of increasing difficulty: [\[Easy\]](#), [\[Intermediate\]](#), and [\[Hard\]](#).

1 Euclidean path integrals and lattice scalar field theory

Question 1. [\[Easy\]](#) What is a Wick rotation, and why does rotating to Euclidean time make the path integral converge? Contrast the real-time weight $e^{iS/\hbar}$ with the Euclidean weight $e^{-S_E/\hbar}$.

Solution. A Wick rotation analytically continues the time variable $t \rightarrow -i\tau$ (equivalently rotating the time contour by 90° in the complex plane). Under it the oscillatory real-time factor $e^{-iHt/\hbar}$ becomes the exponentially damped $e^{-H\tau/\hbar}$. In the path integral the weight changes from the pure phase $e^{iS/\hbar}$, whose rapid oscillations make the integral only conditionally convergent, to the real positive Boltzmann-like weight $e^{-S_E/\hbar}$ with $S_E \geq 0$. The latter is absolutely convergent and can be interpreted as a probability weight, which is what makes Monte Carlo sampling possible. ■

Question 2. [\[Intermediate\]](#) Write down the discretized Euclidean action for a free scalar field on a hypercubic lattice of spacing a , and explain how a acts as an ultraviolet regulator via the Brillouin zone.

Solution.

$$S_E = a^4 \sum_n \left[\frac{1}{2} \sum_\mu \left(\frac{\phi(n + \hat{\mu}) - \phi(n)}{a} \right)^2 + \frac{1}{2} m^2 \phi(n)^2 \right]. \quad (1)$$

On a lattice of spacing a the field lives only at discrete sites, so its Fourier modes are periodic and momenta are restricted to the first Brillouin zone, $p_\mu \in (-\pi/a, \pi/a]$. This provides a hard momentum cutoff $\Lambda \sim \pi/a$: no modes with wavelength shorter than a exist. Loop integrals that would diverge in the continuum become finite sums over the Brillouin zone, so $a > 0$ regulates the UV divergences of the theory. ■

Question 3. [\[Hard\]](#) Derive the Euclidean path integral for a single non-relativistic particle starting from the transition amplitude $\langle x_f | e^{-iH(t_f - t_i)/\hbar} | x_i \rangle$. Show explicitly how the Trotter splitting of $H = p^2/2m + V(x)$ and insertion of complete sets of position and momentum eigenstates produces the discretized action, and state the continuum action recovered in the limit.

Solution. Wick rotate $t \rightarrow -i\tau$ so that $e^{-iH(t_f - t_i)/\hbar} \rightarrow e^{-H(\tau_f - \tau_i)/\hbar}$, and split the interval into N steps of size $\epsilon = (\tau_f - \tau_i)/N$:

$$\langle x_f | e^{-H(\tau_f - \tau_i)/\hbar} | x_i \rangle = \langle x_f | \left(e^{-\epsilon H/\hbar} \right)^N | x_i \rangle. \quad (2)$$

Insert complete sets of position eigenstates $\int dx_k |x_k\rangle\langle x_k|$ between the factors. For each factor use the Trotter approximation $e^{-\epsilon H/\hbar} \approx e^{-\epsilon V/\hbar} e^{-\epsilon p^2/2m\hbar}$ (valid to $\mathcal{O}(\epsilon^2)$), and evaluate the kinetic factor by inserting momentum eigenstates and doing the Gaussian p -integral:

$$\langle x_{k+1} | e^{-\epsilon p^2/2m\hbar} | x_k \rangle = \sqrt{\frac{m}{2\pi\hbar\epsilon}} \exp\left[-\frac{m}{2\hbar\epsilon}(x_{k+1} - x_k)^2\right]. \quad (3)$$

Collecting all factors gives

$$\langle x_f | e^{-H(\tau_f - \tau_i)/\hbar} | x_i \rangle = \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi\hbar\epsilon}\right)^{N/2} \int \prod_{k=1}^{N-1} dx_k \exp\left[-\frac{1}{\hbar} \sum_{k=0}^{N-1} \epsilon \left(\frac{m}{2} \left(\frac{x_{k+1} - x_k}{\epsilon}\right)^2 + V(x_k)\right)\right]. \quad (4)$$

The exponent is $-S_E/\hbar$ with the discretized Euclidean action in brackets. Taking $\epsilon \rightarrow 0$, the sum becomes an integral and one recovers

$$S_E[x] = \int_{\tau_i}^{\tau_f} d\tau \left[\frac{m}{2} \dot{x}^2 + V(x) \right], \quad (5)$$

i.e. the Euclidean action with the sign of the potential term flipped relative to the real-time Lagrangian. ■

2 Gauge theories on a lattice

Question 4. [Easy] Define the link variable $U_\mu(n)$ and state its gauge transformation law. Why is a link variable, rather than the field A_μ directly, the natural object on the lattice?

Solution. The link variable is the parallel transporter along the edge from site n to $n + \hat{\mu}$,

$$U_\mu(n) = \exp(ia g A_\mu(n)) \in SU(N), \quad (6)$$

and it transforms as $U_\mu(n) \rightarrow g(n) U_\mu(n) g(n + \hat{\mu})^\dagger$. It is the natural variable because gauge invariance is realized *exactly* on the lattice through group-valued links: products of links along closed loops are gauge invariant, and one never has to expand in the algebra-valued A_μ . This keeps the lattice measure (the Haar measure over $SU(N)$) compact and gauge invariant to all orders in a . ■

Question 5. [Intermediate] Define the plaquette $U_{\mu\nu}(n)$ and write down the Wilson gauge action. Explain why the plaquette is the smallest gauge-invariant object built from links.

Solution. The plaquette is the ordered product of links around an elementary square:

$$U_{\mu\nu}(n) = U_\mu(n) U_\nu(n + \hat{\mu}) U_\mu^\dagger(n + \hat{\nu}) U_\nu^\dagger(n), \quad (7)$$

and the Wilson gauge action is

$$S_W = \beta \sum_n \sum_{\mu < \nu} \left(1 - \frac{1}{N} \text{Re Tr } U_{\mu\nu}(n)\right), \quad \beta = \frac{2N}{g^2}. \quad (8)$$

A single link is not gauge invariant (it transforms with $g(n)$ on one end and $g(n + \hat{\mu})^\dagger$ on the other). Only a *closed* loop, whose endpoints coincide, has all the intermediate gauge factors cancel, leaving Tr invariant under $U \rightarrow g U g^\dagger$. The plaquette is the shortest such closed loop on a hypercubic lattice, hence the smallest gauge-invariant object. ■

Question 6. [Hard] By expanding the link variables in powers of the lattice spacing a , show that the Wilson gauge action reduces to the continuum Yang–Mills action $\frac{1}{4} \int d^4x F_{\mu\nu}^a F^{a\mu\nu}$ as $a \rightarrow 0$. Identify the leading lattice-artifact correction.

Solution. Write $U_\mu(n) = e^{iagA_\mu(n)}$ and use the Baker–Campbell–Hausdorff formula to combine the four exponentials of the plaquette. The linear terms in A cancel around the closed loop; the surviving terms assemble into the field strength

$$U_{\mu\nu}(n) = \exp\left(ia^2 g F_{\mu\nu}(n) + \mathcal{O}(a^3)\right), \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu]. \quad (9)$$

Then

$$1 - \frac{1}{N} \text{Re Tr } U_{\mu\nu} = \frac{a^4 g^2}{2N} \text{Tr}(F_{\mu\nu} F_{\mu\nu}) + \mathcal{O}(a^6). \quad (10)$$

Multiplying by $\beta = 2N/g^2$ and summing, $a^4 \sum_n \rightarrow \int d^4x$, gives

$$S_W \rightarrow \frac{1}{2} \int d^4x \text{Tr } F_{\mu\nu} F_{\mu\nu} = \frac{1}{4} \int d^4x F_{\mu\nu}^a F^{a\mu\nu}, \quad (11)$$

the continuum Yang–Mills action. The leading correction is $\mathcal{O}(a^2)$ (the a^6 term relative to the a^4 leading term), a dimension-six operator; it is this $\mathcal{O}(a^2)$ artifact that improved actions (e.g. Symanzik improvement) are designed to cancel. ■

3 Fermions on a lattice

Question 7. [Easy] State the fermion doubling problem. How many doublers does the naive discretization of the Dirac operator produce in d dimensions, and where in the Brillouin zone do they sit?

Solution. Replacing $\gamma_\mu \partial_\mu$ by the symmetric difference $\gamma_\mu \frac{\psi(n+\hat{\mu}) - \psi(n-\hat{\mu})}{2a}$ gives an inverse propagator $\propto \frac{i}{a} \sum_\mu \gamma_\mu \sin(p_\mu a)$, which vanishes not only at $p_\mu = 0$ but at every $p_\mu \in \{0, \pi/a\}$. Each momentum component thus contributes two zeros, giving 2^d degenerate species; in $d = 4$ this is 16 fermions where the theory should describe one. The extra 15 doublers sit at the corners and edge-centers of the Brillouin zone (momentum components equal to π/a). ■

Question 8. [Intermediate] Compare Wilson and staggered fermions. What symmetry does each approach sacrifice or preserve, and by what mechanism does each remove (or reduce) the doublers?

Solution. Wilson fermions add a dimension-five term $-\frac{ar}{2} \sum_\mu \Delta_\mu \psi$ (a discretized Laplacian) that gives each doubler a mass $\sim 1/a$, so the 15 doublers decouple as $a \rightarrow 0$ and only the physical mode survives. The price is that this term explicitly breaks chiral symmetry at finite a , requiring an additive mass renormalization / fine-tuning to reach the chiral limit. Staggered (Kogut–Susskind) fermions instead spread the four Dirac components over the 2^4 corners of a hypercube (spin diagonalization), reducing the 16 species to 4 "tastes." This preserves a remnant $U(1)$ chiral symmetry at finite a (protecting the mass), but the residual four-fold taste degeneracy must be removed (e.g. by the fourth-root trick), which introduces taste-breaking lattice artifacts. ■

Question 9. [Hard] State the Nielsen–Ninomiya no-go theorem and its assumptions. Explain how the Ginsparg–Wilson relation evades it and what form of lattice chiral symmetry it permits.

Solution. The Nielsen–Ninomiya theorem states that a lattice Dirac operator cannot simultaneously be (i) local, (ii) translation invariant, (iii) chirally symmetric in the naive sense $\{D, \gamma_5\} = 0$, and (iv) free of doublers (having the correct continuum limit). Any operator satisfying (i), (ii), (iv) must break exact chiral symmetry. The Ginsparg–Wilson relation

$$\{D, \gamma_5\} = a D \gamma_5 D \quad (12)$$

evades the theorem by relaxing exact anticommutation: the right-hand side is an $\mathcal{O}(a)$ term that vanishes in the continuum. A D satisfying it (e.g. the overlap operator) is local and doubler-free, and is invariant under the Lüscher lattice chiral transformation $\delta\psi = \gamma_5(1 - \frac{a}{2}D)\psi$, $\delta\bar{\psi} = \bar{\psi}(1 - \frac{a}{2}D)\gamma_5$. This is an exact, modified chiral symmetry on the lattice that reproduces the correct chiral anomaly and index theorem, circumventing the no-go obstruction without reintroducing doublers. ■

4 Continuum limit

Question 10. [Easy] *What is the physical correlation length ξ_{phys} in terms of the lattice spacing a and the dimensionless correlation length ξ ? Why must $\xi \rightarrow \infty$ (in lattice units) to reach the continuum?*

Solution. The physical correlation length is $\xi_{\text{phys}} = a\xi$, where ξ is measured in lattice units (the number of lattice sites over which correlations decay). Equivalently $\xi = 1/(am_{\text{phys}})$ for a physical mass m_{phys} . To take $a \rightarrow 0$ while keeping ξ_{phys} (a physical quantity) fixed, the dimensionless $\xi = \xi_{\text{phys}}/a$ must diverge. A diverging correlation length in lattice units is precisely the hallmark of a critical point, so the continuum limit lives at a critical point of the lattice model. ■

Question 11. [Intermediate] *Explain why the continuum limit must be taken at a second-order (critical) point of the lattice theory, and why a first-order transition is unsuitable.*

Solution. At a second-order phase transition the correlation length diverges *continuously* and there is no latent heat, so one can tune the bare couplings to make $\xi \rightarrow \infty$ smoothly; this is what lets $a = \xi_{\text{phys}}/\xi \rightarrow 0$ while physics stays fixed, and it also guarantees loss of memory of the lattice (universality). At a first-order transition the correlation length stays *finite* across the transition (two phases coexist with a discontinuity and latent heat), so ξ never diverges and a cannot be sent to zero — there is no scaling region in which lattice artifacts vanish. Hence only second-order critical points define a continuum limit. ■

Question 12. [Hard] *Using asymptotic freedom and the two-loop beta function of $SU(N)$ gauge theory, derive how the lattice spacing a depends on the bare coupling g near the continuum limit. Explain what "asymptotic scaling" means and how it is used to set the lattice scale.*

Solution. The bare coupling runs with the cutoff $\Lambda = 1/a$ according to

$$-a \frac{dg}{da} = \beta(g) = -\beta_0 g^3 - \beta_1 g^5 - \dots, \quad \beta_0 = \frac{1}{(4\pi)^2} \frac{11N}{3}, \quad \beta_1 = \frac{1}{(4\pi)^4} \frac{34}{3} N^2. \quad (13)$$

Integrating the two-loop equation gives the lattice spacing in terms of g :

$$a \Lambda_L = (\beta_0 g^2)^{-\beta_1/2\beta_0^2} \exp\left(-\frac{1}{2\beta_0 g^2}\right) [1 + \mathcal{O}(g^2)], \quad (14)$$

where Λ_L is the lattice scale (a renormalization-group invariant integration constant). Since $\beta_0 > 0$, $a \rightarrow 0$ as $g \rightarrow 0$: the continuum limit is the weak-coupling limit, i.e. asymptotic freedom. "Asymptotic scaling" is the regime where this two-loop formula holds, so that every dimensionful lattice quantity behaves as (its value in lattice units) $\times a(g)$ with the same exponential dependence on g ; observing that dimensionless ratios become g -independent confirms one is in the scaling window. In practice one measures a known physical quantity (e.g. a hadron mass or the string tension) in lattice units at a given g and uses $a(g)$ to convert all other lattice results to physical units. ■

5 Chemical potential and the numerical sign problem

Question 13. [Easy] *How is a quark chemical potential μ introduced into the lattice action, and why can it not simply multiply the number operator naively? State qualitatively why $\mu \neq 0$ leads to a complex action.*

Solution. A chemical potential couples to conserved quark number. On the lattice a naive term $\mu \bar{\psi} \gamma_0 \psi$ produces μ^2/a^2 divergences in the continuum limit. The cure (Hasenfratz–Karsch) is to introduce μ like an imaginary constant temporal gauge field, multiplying the forward and backward temporal hopping terms by $e^{+\mu a}$ and $e^{-\mu a}$ respectively. This exponential (link-like) coupling is finite in the continuum. Because the two temporal directions are weighted asymmetrically ($e^{+\mu a} \neq e^{-\mu a}$ for real μ), the Dirac operator loses its γ_5 -Hermiticity and its determinant becomes complex. ■

Question 14. [Intermediate] *Explain the origin of the sign problem in terms of γ_5 -Hermiticity of the Dirac operator. Why does reweighting from a positive ensemble fail as the volume grows?*

Solution. At $\mu = 0$ the Dirac operator obeys γ_5 -Hermiticity, $M^\dagger = \gamma_5 M \gamma_5$, which forces $\det M$ to be real (and, for degenerate pairs, positive), so it can serve as a probability weight. The finite-density coupling $e^{\pm \mu a}$ breaks this relation: instead $M(\mu)^\dagger = \gamma_5 M(-\mu^*) \gamma_5$, so for real μ the determinant is genuinely complex. Importance sampling then has no positive weight to sample from. One can reweight, writing observables as ratios of averages over a positive ensemble (e.g. phase-quenched), but the reweighting factor is the fluctuating phase $e^{i\theta}$ whose average $\langle e^{i\theta} \rangle \sim e^{-V\Delta f}$ is exponentially small in the volume. The signal-to-noise ratio therefore degrades exponentially with V , requiring an exponentially large number of configurations — the sign problem. ■

Question 15. [Hard] *Quantify the severity of the sign problem via the expectation value of the phase $\langle e^{i\theta} \rangle_{pq}$ in the phase-quenched ensemble, and show that it is suppressed exponentially in the volume, $\langle e^{i\theta} \rangle \sim e^{-V\Delta f}$. Relate Δf to a free-energy difference and explain the resulting cost of a reweighted Monte Carlo estimate.*

Solution. Split the complex weight as $\det M(\mu) = |\det M(\mu)| e^{i\theta}$. The full and phase-quenched partition functions are $Z = \int \mathcal{D}U |\det M| e^{i\theta} e^{-S_g}$ and $Z_{pq} = \int \mathcal{D}U |\det M| e^{-S_g}$. Their ratio is exactly the average phase in the phase-quenched ensemble:

$$\langle e^{i\theta} \rangle_{pq} = \frac{Z}{Z_{pq}} = e^{-V(f-f_{pq})} \equiv e^{-V\Delta f}, \quad (15)$$

where f and f_{pq} are the free-energy densities of the full and phase-quenched theories and $\Delta f = f - f_{pq} \geq 0$. Since Δf is generically nonzero (the phase-quenched theory has different physics — e.g. a pion condensate — so its free energy differs), the average phase is exponentially small in

the volume. Reweighting estimates $\langle O \rangle = \langle O e^{i\theta} \rangle_{pq} / \langle e^{i\theta} \rangle_{pq}$; the statistical error of the denominator relative to its value grows like $1/(\sqrt{N_{\text{cfg}}} \langle e^{i\theta} \rangle_{pq})$. To keep it fixed one needs

$$N_{\text{cfg}} \gtrsim \langle e^{i\theta} \rangle_{pq}^{-2} \sim e^{2V\Delta f}, \quad (16)$$

i.e. an exponentially large number of configurations in the volume — the quantitative statement of the sign problem's severity. ■

6 Techniques to circumvent the sign problem

Question 16. [Easy] *Distinguish sign quenching from phase quenching. Why is the phase-quenched theory generally not the theory one wants to solve?*

Solution. Sign quenching replaces the (real but possibly negative) weight by its absolute value, discarding the sign; phase quenching replaces the complex weight $\det M = |\det M| e^{i\theta}$ by its modulus $|\det M|$, discarding the phase. In QCD phase quenching corresponds to giving the anti-quark an opposite chemical potential, i.e. simulating an isospin chemical potential rather than a baryon one. That theory has genuinely different physics (e.g. a charged pion-condensation transition at $\mu \approx m_\pi/2$) and hence a different free energy, so it does not reproduce the finite-baryon-density results one actually wants — the discarded phase carries the essential physics. ■

Question 17. [Intermediate] *Describe two conventional finite-density approaches (e.g. Taylor expansion in μ/T , imaginary chemical potential with analytic continuation, or reweighting). For each, state its domain of validity and its main limitation.*

Solution. *Taylor expansion:* simulate at $\mu = 0$ (where the weight is positive) and expand observables in powers of μ/T , computing the Taylor coefficients as $\mu = 0$ correlators. Valid for small μ/T ; limited by the radius of convergence (set by the nearest singularity, e.g. a critical endpoint) and by the rapidly growing statistical noise of high-order coefficients. *Imaginary chemical potential:* for $\mu = i\mu_I$ the determinant is real and positive, so one simulates directly and then analytically continues results back to real μ . Valid within the first Roberge–Weiss sector $\mu_I/T < \pi/3$; limited by the ansatz used for analytic continuation and by the RW transition bounding the accessible region. (*Reweighting*, a third option, is exact in principle but suffers the exponential overlap/sign degradation discussed above.) All three are controlled only at small chemical potential and cannot reach the cold, dense region of the phase diagram. ■

Question 18. [Hard] *Explain the idea of Lefschetz thimbles: how deforming the integration contour into complexified field space onto steepest-descent manifolds tames the sign problem. What new difficulties (e.g. residual phase, multiple thimbles, Jacobian) does the method introduce?*

Solution. The integrand e^{-S} with complex action S is holomorphic, so by Cauchy's theorem the integration contour over real fields can be deformed into complexified field space without changing the integral. One deforms it onto a union of *Lefschetz thimbles* — the steepest-descent (unstable) manifolds attached to the critical points $\partial S = 0$, defined by the holomorphic gradient flow $\dot{z} = \overline{\partial S / \partial z}$. On a thimble the imaginary part $\text{Im } S$ is *constant* (equal to its value at the critical point), so the rapidly oscillating phase that caused the sign problem is factored out and $e^{-\text{Re } S}$ provides a real, positive sampling weight. The new difficulties are: (i) a *residual phase* from the orientation of the thimble — the Jacobian of the tangent-space frame (the induced measure) is itself complex and

must be reweighted, though its fluctuations are milder; (ii) *multiple thimbles* — several critical points may contribute, weighted by integer intersection numbers, and combining them coherently can reintroduce cancellations (a "global" sign problem); (iii) the practical cost of *integrating the gradient flow* and computing the Jacobian (det of the flow) at every step, which is numerically expensive; and (iv) subtleties near *Stokes phenomena*, where the set of contributing thimbles changes discontinuously as parameters vary. ■